

Aziz Laadhar

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EDUCATION

EPFL — M.Sc. Data Science, Minor in Financial Engineering Sep 2023 – Present
Coursework: Stochastic Calculus, Econometrics, Portfolio Optimization, Quantitative Risk

EPFL — B.Sc. Communication Systems Sep 2020 – Jul 2023
Teaching Assistant: Operating Systems, Computer Architecture, Computer Networks.

QUANTITATIVE RESEARCH EXPERIENCE

Sirius Energy SA Lugano, Switzerland
Quantitative Research Analyst — Power and Gas Trading Jul 2024 – Feb 2025

- Developed an autoregressive supply/demand-curve model to forecast day-ahead power markets from 25M+ bids on a 500-point price grid; estimated curve operators as a 750k-variable SOCP in MOSEK Fusion.
- Converted forecast curves into clearing-price forecasts using the desk's EUPHEMIA market-coupling replica; reduced OOS hourly MAE from (€12.4 to €9.2/MWh) versus the desk's model, with 79% lower maximum absolute error.
- Reverse-engineered Italy's undocumented D+1 import ceiling using a two-stage random-forest hurdle model with 42 features; rolling-window validation: F1 0.90, conditional R^2 0.89 vs. 0.72 for OLS; deployed daily to traders.
- Built a counterfactual day-ahead market-clearing backtest that re-solved the market under alternative inputs to attribute zonal price and net-position errors to offer curves, block orders, and transfer capacities.
- Labeled the generation technology of 34% of 658k unlabeled power-market offers via a MILP over 743 hours; identified latent PV supply from residual offers that tracked solar generation and concentrated at the low-price end of the stack.

RESEARCH EXPERIENCE

Harvard University Cambridge, MA
Research Scholar — Advisor: Prof. Jia Liu, Prof. Na Li Feb 2026 – Present

- Built self-supervised representation-learning components for high-dimensional spatiotemporal data, compressing multi-channel video into low-dimensional latent trajectories for behavioral-state inference.
- Designed a 65-task benchmark for quantitative video inference, calibrating scores to measurement noise and human error; nominal 95% intervals covered 29% of 66 targets, while confidence calibration reached ECE 0.32 across 1,872 predictions.

SLB — Schlumberger-Doll Research Cambridge, MA
Machine Learning Researcher — Mathematical Physics and Modeling Jul 2025 – Feb 2026

- Built a physics-constrained neural model for 2D resistivity inversion, embedding the forward simulator as a data-consistency penalty; reduced median MSE 0.081→0.070 vs. CNN baseline, with one forward pass vs. tens-hundreds of iterative solver steps.
- Developed structure-preserving losses and adaptive sampling for noisy inverse problems, reducing PDE residuals 35%; stress-tested robustness under noise and distribution shift through systematic ablations.

EPFL Visual Intelligence for Transportation Laboratory Lausanne, Switzerland
Research Assistant — Advisor: Prof. Alexandre Alahi Sep 2022 – Jun 2023

- Trained a graph-transformer policy for the constrained Dial-a-Ride Problem by imitation on 50k Gurobi solutions, achieving a 2.67% cost gap to exact optimization on 100 held-out instances.
- Trained feasibility-masked PPO, reducing ride-time violations 1.61→0.33; distilled the policy to 1.55 s/instance while maintaining a 2.37% cost gap.

SELECTED PROJECTS

Mercuria Campus Power Trading Workshop (EPFL) — Winning Team Apr 2024

- Priced electricity caps across four Australian NEM regions using Monte Carlo with a mean-reverting jump-diffusion model and two Poisson jump processes; calibrated by MLE on 750k half-hourly prices.

Optiver EPFL Hackathon — Top 3 Team Dec 2023

- Built regression-based Tesla price/volatility signals and iterated feature, model, and trading-rule selection against simulated PnL.

TECHNICAL SKILLS

Programming: Python (NumPy, SciPy, pandas, scikit-learn, PyTorch), C++, SQL, R; Linux, Git.

Quantitative Methods: Time-series forecasting, state-space models, Kalman filtering, ARIMA, functional autoregression, Monte Carlo, MLE, constrained optimization, market microstructure, backtesting.

Machine Learning: Tree ensembles, deep learning, physics-informed ML, graph neural networks, self-supervised learning, deep RL.

Languages: French and Arabic (native); English (C2).